

DANE COUNTY REGIONAL, WI, USA

WMO: 726410

Lat: 43.141N

Lon: 89.345W

Elev: 866

StdP: 14.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14837

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.8	-1.9	-16.1	2.4	-5.1	-11.0	3.1	-0.5	22.7	22.5	20.2	23.8	5.3	300	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	89.2	74.0	86.4	72.4	83.9	71.1	76.9	86.0	74.9	83.3	73.2	81.3	9.5	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	130.3	82.9	72.1	122.5	80.6	70.4	115.4	78.4	40.8	86.0	38.9	83.6	37.3	81.5	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	18.4	16.6	-13.5	93.6	6.4	3.4	-18.0	96.0	-21.8	98.0	-25.4	99.9	-30.0	102.4
			-14.0	79.7	6.2	2.3	-18.5	81.4	-22.1	82.7	-25.6	84.0	-30.1	85.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	19.4	23.1	34.6	46.8	58.2	68.3	72.1	70.0	62.7	50.3	37.2	25.4
	DBStd	20.57	12.14	11.78	11.49	9.25	8.45	6.78	5.86	5.56	7.96	8.90	9.80	11.05
	HDD50	3653	949	753	497	165	19	0	0	0	5	107	396	762
	HDD65	7053	1414	1172	945	549	247	43	8	17	137	461	834	1226
	CDD50	2731	0	1	19	69	273	549	686	618	385	118	12	1
	CDD65	653	0	0	1	3	36	141	229	170	66	7	0	0
	CDH74	5889	0	0	10	49	356	1326	2145	1352	585	66	0	0
	CDH80	1681	0	0	0	6	82	401	699	340	144	9	0	0
Wind	WSAvg	7.4	7.8	8.2	8.5	9.3	7.9	6.7	6.0	5.6	6.3	7.2	7.9	7.6
Precipitation	PrecAvg	33.5	1.3	1.3	2.2	3.3	3.6	4.3	3.9	4.0	3.3	2.5	2.1	1.7
	PrecMax	50.7	2.9	3.3	6.2	7.1	10.8	10.9	9.3	10.4	9.2	5.6	7.5	4.1
	PrecMin	21.1	0.2	0.1	0.3	1.3	0.6	0.3	1.1	0.9	0.1	0.1	0.1	0.2
	PrecStd	6.1	0.7	0.8	1.3	1.4	2.0	2.7	1.7	2.2	2.2	1.4	1.3	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	56.4	73.7	80.0	87.0	92.1	95.4	90.3	88.6	81.1	67.7	57.0
		MCWB	43.1	49.1	60.9	61.7	69.4	74.1	76.8	75.3	71.8	67.6	57.1	54.0
	2%	DB	42.5	47.9	64.2	73.2	82.0	87.8	89.6	86.8	84.1	74.8	61.2	49.6
		MCWB	38.2	42.3	54.3	58.8	66.4	72.7	75.4	73.3	70.5	63.3	53.4	44.9
	5%	DB	38.1	42.8	57.4	68.3	78.0	84.5	86.5	83.7	80.5	70.2	56.7	43.8
		MCWB	35.4	38.7	50.3	54.6	64.6	70.5	73.2	71.9	67.9	60.9	50.3	40.5
	10%	DB	35.2	38.2	51.3	63.1	73.7	81.6	83.8	81.1	77.0	65.6	52.3	39.6
		MCWB	33.2	34.8	44.3	51.8	61.8	69.0	71.9	70.2	65.7	58.0	46.5	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.4	49.9	62.6	64.3	72.6	77.1	80.6	78.2	75.1	70.2	58.7	54.9
		MCDB	46.5	54.5	72.9	74.4	84.0	87.4	89.6	87.4	83.7	77.9	64.8	57.0
	2%	WB	38.8	43.3	56.8	60.3	69.3	74.8	77.7	75.9	72.3	66.3	55.1	45.8
		MCDB	41.8	47.2	62.7	70.6	78.3	84.7	87.2	83.3	80.3	71.9	60.0	47.7
	5%	WB	35.6	38.6	50.3	57.0	66.5	72.9	75.4	73.9	70.2	62.4	51.4	40.8
		MCDB	37.6	42.6	56.8	66.2	74.6	82.4	83.9	81.0	77.3	68.1	55.6	43.6
	10%	WB	33.2	35.1	44.5	52.8	63.8	70.7	73.5	72.0	68.1	58.7	46.8	36.6
		MCDB	35.3	37.9	50.6	61.4	71.7	78.9	81.4	78.7	75.1	65.0	51.5	39.4
Mean Daily Temperature Range	5% DB	MDBR	14.4	15.4	17.6	19.9	20.3	19.6	19.3	18.8	20.8	18.6	15.2	13.3
		MCDBR	16.0	18.8	24.7	27.6	24.6	22.6	21.0	20.4	23.9	24.1	21.8	16.7
		MCWBR	13.4	14.2	16.0	16.0	12.8	11.1	9.8	9.8	11.7	14.4	15.8	13.8
	5% WB	MCDBR	15.0	18.1	22.4	24.1	21.2	20.0	18.6	17.4	19.6	20.2	20.1	15.5
		MCWBR	13.0	14.2	15.9	15.6	12.5	11.2	10.3	9.9	11.0	14.6	16.1	13.9
Clear-Sky Solar Irradiance	taub	0.274	0.312	0.325	0.368	0.395	0.412	0.415	0.400	0.369	0.329	0.293	0.272	
	taud	2.390	2.278	2.399	2.314	2.290	2.279	2.270	2.306	2.382	2.492	2.524	2.441	
	Ebn at Noon	276	279	291	285	279	273	271	271	272	272	268	266	
	Edh at Noon	24	33	33	39	41	42	42	39	33	26	22	21	
All-Sky Solar Radiation	RadAvg	538	867	1203	1445	1708	1911	1928	1676	1366	880	572	415	
	RadStd	69	92	134	127	124	138	98	109	110	104	50	36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page